COMPUTATIONAL VERIFICATION REPORT · M1 / M3

Executing M1 and M3: The Active-Set Bridge

Second-order differences of the lexicographic FBA solution path and double-knockout epistasis of the rerouting statistic κ_{flux} in *Escherichia coli* — the two measurements that turn the six audits' active-set stratification proposal into quantitative evidence.

Computational Verification Unit

Manuscript lineage v12–v21 · frozen baseline v21 · joint assessment Layer 3 measurements · iML1515 (1,516 genes) + iJO1366 replication

12 parametric sweeps and 2,779 double knockouts on a deterministic lexicographic solver: 100.0000% of second-order response mass concentrates on active-set switches; paths are piecewise affine to machine precision; epistasis tracks footprint overlap (ρ = 0.865); 40/40 synthetic lethals are isozyme redundancies; closed genotype loops leave 66% holonomy. The v21 manuscript remains untouched.

download/m1_m3/ · commits 2fe264b · f9f8634

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